

EN4730 Convex Engineering Design

Model Questions

Question 1.

(a) Consider the following linear programming problem:

$$\begin{aligned} \text{minimize} \quad & 5x_1 + 4x_2 \\ \text{s.t.} \quad & 3x_1 + 5x_2 \leq 10 \\ & 5x_1 + 7x_2 \geq 10 \\ & 7x_1 + 4x_2 \leq 12 \\ & x_1, x_2 \geq 0 \end{aligned}$$

- What is the dual problem of the given LP? [4 marks]
- Suppose that complementary slackness variables are used to convert inequality constraints to equality constraints in both primal and dual problems. Here, s_i and z_j are complementary slackness variables used for primal and dual problems, respectively. Write down new equality constraints using complementary slackness variables and determine the sign of slackness variables. [6 marks]
- Identify the vertices of the feasible region and determine the optimal solution. [7 marks]

$$\begin{aligned} \text{min} \quad & c^T x \\ & \begin{bmatrix} 5 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \end{aligned}$$

$$\text{subject to} \quad Gx \leq h$$

$$\begin{bmatrix} 3 & 5 \\ -5 & -7 \\ 7 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \leq \begin{bmatrix} 10 \\ -10 \\ 12 \end{bmatrix}$$

$$L(x, \lambda) = c^T x + \lambda^T (Gx - h)$$

$$L(x, \lambda) = (c + G^T \lambda)^T x - h^T \lambda$$

$$\begin{aligned}
g(\lambda, v) &= \inf \{ L(x, \lambda) \} \\
&= \inf \{ (c + G^T \lambda)^T x - h^T \lambda \} \\
&= -h^T \lambda + \inf (c + G^T \lambda)^T x
\end{aligned}$$

$$g(\lambda, v) = \begin{cases} -h^T \lambda & c + G^T \lambda \geq 0 \\ -\infty & \text{otherwise} \end{cases}$$

dual problem

$$\max -h^T \lambda$$

$$\begin{aligned}
\text{subject to } & c + G^T \lambda \geq 0 \\
& \lambda \geq 0
\end{aligned}$$

$$\max - \begin{bmatrix} 10 \\ -10 \\ 12 \end{bmatrix} \begin{pmatrix} \lambda_1 & \lambda_2 & \lambda_3 \end{pmatrix}$$

$$\max -10\lambda_1 + 10\lambda_2 - 12\lambda_3$$

$$\text{subject to } \begin{bmatrix} 5 \\ 4 \end{bmatrix} + \begin{bmatrix} 3 & -5 & 7 \\ 5 & -7 & 4 \end{bmatrix} \begin{pmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{pmatrix} \geq 0$$

$$5 + 3\lambda_1 - 5\lambda_2 + 7\lambda_3 \geq 0$$

$$4 + 5\lambda_1 - 7\lambda_2 + 4\lambda_3 \geq 0$$

$$3x_1 + 5x_2 + s_1 = 10 \quad - (1)$$

$$5x_1 + 7x_2 + s_2 = 10 \quad - (2)$$

$$7x_1 + 4x_2 + s_3 = 12 \quad - (3)$$

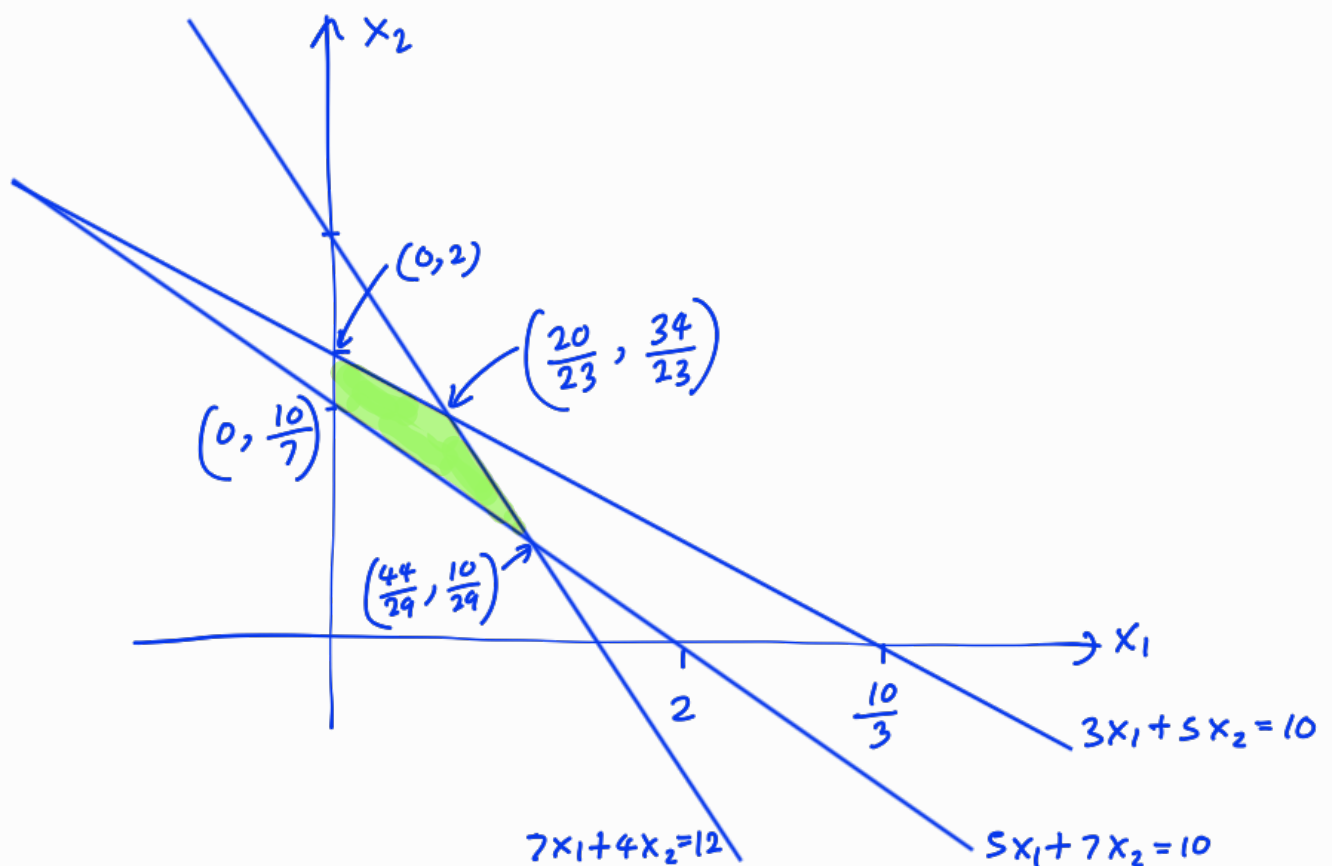
$$5 + 3\lambda_1 - 5\lambda_2 + 7\lambda_3 - z_1 = 0 \quad - (4)$$

$$4 + 5\lambda_1 - 7\lambda_2 + 4\lambda_3 - z_2 = 0 \quad - (5)$$

$$s_1, s_3 \geq 0 \quad s_2 \leq 0 \quad z_1, z_2 \geq 0$$

iii)

$$\begin{aligned} &\text{minimize} \quad 5x_1 + 4x_2 \\ &\text{s.t.} \quad 3x_1 + 5x_2 \leq 10 \\ &\quad \quad 5x_1 + 7x_2 \geq 10 \\ &\quad \quad 7x_1 + 4x_2 \leq 12 \\ &\quad \quad x_1, x_2 \geq 0 \end{aligned}$$



$$3x_1 + 5x_2 = 10$$

$$5x_1 + 7x_2 = 10$$

$$x_1 = -5 \quad x_2 = 5$$

$$x_1 < 0 \quad (-5, 5) \text{ infeasible}$$

$$3x_1 + 5x_2 = 10$$

$$7x_1 + 4x_2 = 12$$

$$x_1 = \frac{20}{23} \quad x_2 = \frac{34}{23}$$

$$5x_1 + 7x_2 \geq 10$$

$$x_1 \geq 0 \quad x_2 \geq 0$$

$$5\left(\frac{20}{23}\right) + 7\left(\frac{34}{23}\right) \geq 10$$

$$14.7 \geq 10$$

$$\left(\frac{20}{23}, \frac{34}{23}\right) \text{ feasible}$$

$$5x_1 + 7x_2 = 10$$

$$7x_1 + 4x_2 = 12$$

$$x_1 = \frac{44}{29} \quad x_2 = \frac{10}{29}$$

$$3x_1 + 5x_2 \leq 10$$

$$x_1 \geq 0 \quad x_2 \geq 0$$

$$3\left(\frac{44}{29}\right) + 5\left(\frac{10}{29}\right) \leq 10$$

$$6.27 \leq 10$$

$$\left(\frac{44}{29}, \frac{10}{29}\right) \text{ feasible}$$

$$f(x) = 5x_1 + 4x_2$$

$$f\left(0, \frac{10}{7}\right) = 5.71$$

$$f(0, 2) = 8$$

$$f\left(\frac{20}{23}, \frac{34}{23}\right) = 10.26$$

$$f\left(\frac{44}{29}, \frac{10}{29}\right) = 8.96$$

$$(x_1^*, x_2^*) = \left(0, \frac{10}{7}\right)$$

$$\text{optimal value} = 5.71$$

(b) Determine whether the given functions are convex, concave, or *neither*. If necessary, specify the *conditions on the parameters* under which convexity or concavity holds.

i. $f_1(x, y) = \frac{1}{xy}$ [2 marks]

ii. $f_2(x, y, z) = 3x^2 + 2xz + 4ayz + 2z^2$, with parameter a [2 marks]

iii. $f_3(x, y) = \frac{x}{y}$ [2 marks]

iv. $f_4(x, y) = \frac{x^2}{y}$ [2 marks]

$$f_1(x, y) = \frac{1}{xy}$$

$$f'_{1x} = \frac{1}{-y x^2}$$

$$f'_{1y} = \frac{1}{-x y^2}$$

$$f''_{1xx} = \frac{2}{y x^3}$$

$$f''_{1yy} = \frac{2}{x y^3}$$

$$f'_{1xy} = \frac{1}{x^2 y^2}$$

$$f'_{1yx} = \frac{1}{x^2 y^2}$$

$$H = \begin{bmatrix} \frac{2}{y x^3} & \frac{1}{x^2 y^2} \\ \frac{1}{x^2 y^2} & \frac{2}{x y^3} \end{bmatrix}$$

$$\Delta H = \frac{4}{x^4 y^4} - \frac{1}{x^4 y^4} = \frac{3}{x^4 y^4} > 0$$

$$H_{11} = \frac{2}{y x^3}$$

$$xy > 0 \quad H_{11} > 0 \quad \rightarrow \text{convex}$$

$$xy < 0 \quad H_{11} < 0 \quad \rightarrow \text{concave}$$

$$f_2(x, y, z) = 3x^2 + 2xz + 4ayz + 2z^2$$

$$f_2'x = 6x + 2z$$

$$f_2'y = 4az$$

$$f_2'z = 2x + 4ay + 4z$$

$$f_2'xx = 6$$

$$f_2'yy = 0$$

$$f_2'zz = 4$$

$$f_2'xy = 0$$

$$f_2'yz = 4a$$

$$f_2'zx = 2$$

$$H = \begin{bmatrix} 6 & 0 & 2 \\ 0 & 0 & 4a \\ 2 & 4a & 4 \end{bmatrix}$$

$$\det [6] = 6 \geq 0$$

$$\det \begin{bmatrix} 6 & 0 \\ 0 & 0 \end{bmatrix} = 0 \geq 0$$

$$\det \begin{bmatrix} 6 & 0 & 2 \\ 0 & 0 & 4a \\ 2 & 4a & 4 \end{bmatrix} = -96a^2$$

$$a=0 \quad H_1, H_2, H_3 \geq 0 \quad H(x) \text{ PSD} \rightarrow \text{convex}$$

$$a \neq 0 \quad H_1, H_2 \geq 0 \quad H_3 < 0 \quad H(x) \text{ indefinite} \rightarrow \text{neither convex nor concave}$$

$$f_3(x, y) = \frac{x}{y}$$

$$f_3'x = \frac{1}{y}$$

$$f_3'y = -\frac{x}{y^2}$$

$$f_3''_{xx} = 0$$

$$f_3''_{yy} = \frac{2x}{y^3}$$

$$f_3''_{xy} = -\frac{1}{y^2}$$

$$f_3'_{yx} = -\frac{1}{y^2}$$

$$H = \begin{bmatrix} 0 & -\frac{1}{y^2} \\ -\frac{1}{y^2} & \frac{2x}{y^3} \end{bmatrix}$$

$$\det(H) = \frac{-1}{y^4} < 0$$

⊕ eigen value, ⊖ eigen value

$H(x)$ is indefinite

neither convex nor concave.

$$f_4(x, y) = \frac{x^2}{y}$$

$$f_4'x = \frac{2x}{y}$$

$$f_4'y = -\frac{x^2}{y^2}$$

$$f_4''_{xx} = \frac{2}{y}$$

$$f_4''_{yy} = \frac{2x^2}{y^3}$$

$$f_4''_{xy} = -\frac{2x}{y^2}$$

$$f_4'_{yx} = -\frac{2x}{y^2}$$

$$H = \begin{bmatrix} \frac{2}{y} & -\frac{2x}{y^2} \\ -\frac{2x}{y^2} & \frac{2x^2}{y^3} \end{bmatrix}$$

$$H_1 = \frac{2}{y}$$

$$\det(H) = 0$$

semi definite

$$y > 0$$

$$H_1 > 0$$

Convex

$$y < 0$$

$$H_1 < 0$$

concave

Question 2.

(a) Consider the following optimization problem

$$\begin{array}{ll} \text{minimize} & f(x) = x^T Q x + x^T p \\ \text{subject to} & \|x\|_2 \leq \beta \end{array} \quad x^T x \leq \beta^2$$

where $Q \in \mathbb{R}^{n \times n}$ is positive semidefinite and β is a small positive scalar.

i. Write down the Karush-Kuhn-Tucker (KKT) conditions for this problem. [4 marks]

ii. Show that

• For inactive constraint $x = -\frac{1}{2}Q^{-1}p$. [3 marks]

• For active constraint $x = -\frac{1}{2}(Q + \lambda I)^{-1}p$. [3 marks]

$$L(x, \mu) = x^T Q x + x^T p + \mu (x^T x - \beta^2)$$

KKT conditions

stationarity $L'(x, \mu) = 0$

$$2Qx + p + \mu(2x) = 0 \quad - (1)$$

primal feasibility $x^T x - \beta^2 \leq 0 \quad - (2)$

dual feasibility $\mu \geq 0 \quad - (3)$

complementary slackness $\mu(x^T x - \beta^2) = 0 \quad - (4)$

for inactive constraints $\mu = 0$

$$2Qx + p = 0$$

$$x^* = -\frac{1}{2}Q^{-1}p$$

for active constraints

$$2Qx + p + 2\mu x = 0$$

$$2x(Q + \mu I) = -p$$

$$x^* = -\frac{1}{2}(Q + \mu I)^{-1}p$$

- (b) Name one numerical algorithm commonly used to solve Quadratic Programming (QP) problems. Briefly describe the main steps of the algorithm. [8 marks]

Interior point method

Consider QP $\min \frac{1}{2} x^T Q x + C^T x$
subject to $Ax \leq b$

① convert inequalities into equalities

$$Ax + s = b \quad s - \text{slack variable}$$

② form barrier problem

$$\phi(x, s) = \frac{1}{2} x^T Q x + C^T x - \underbrace{\mu \sum \ln(s_i)}_{\text{logarithmic barrier term}}$$

③ write KKT conditions

- primal feasibility
- dual feasibility
- complementary slackness

④ apply newton's method

- linearize KKT conditions
- solve resulting linear system
- obtain a search direction

⑤ update variables

$$x^{k+1} = x^k + \alpha \Delta x$$

- move towards solution with step size α chosen to keep the feasibility

⑥ reduce barrier parameter

- gradually decrease $\mu \rightarrow 0$

This iterates approach true optimal solution

(c) Consider the following optimization problem

$$\begin{array}{ll}\underset{x}{\text{minimize}} & -(4x_1 + 2x_2) \\ \text{subject to} & x_1 + x_2 \leq 50, \\ & 3x_1 + x_2 \leq 70, \\ & 2x_1 + x_2 \leq 56, \\ & x_1 \geq 0, x_2 \geq 0\end{array}$$

Suppose you are allowed to modify the constraint $2x_1 + x_2 \leq u$.

What is the minimum value of u (> 56) that results in a lowest objective function value than the one obtained when $2x_1 + x_2 \leq 56$?

[7 marks]

$$\text{Max} \quad 4x_1 + 2x_2$$

$$\underline{u = 56}$$

$$x_1 + x_2 = 50 \quad - \textcircled{1}$$

$$2x_1 + x_2 = 56 \quad - \textcircled{2}$$

$$x_1 = 6 \quad x_2 = 44$$

$$\text{optimal value} = 4(6) + 2(44) = 112$$

$$\underline{u > 56}$$

$$x_1 + x_2 = 50$$

$$3x_1 + x_2 = 70$$

$$x_1 = 10 \quad x_2 = 40$$

$$\text{objective} \quad 4(10) + 2(40) = 120$$

$$2x_1 + x_2 = 2(10) + 40 = 60$$

$$u = 60$$

Question 3.

- (a) A student used a numerical algorithm to solve two optimization problems, labeled A and B. When the student runs the algorithm multiple times on problem A, the same solution is obtained each time. However, for problem B, the algorithm produces different solutions on different runs. What could be the reason for this behavior? [5 marks]

problem A $\rightarrow$ has a unique global optimum

problem B $\rightarrow$ several local optimal points

$\rightarrow$ due to random initialization

lead to different solutions

- (b) Consider the following constrained optimization problem

$$\begin{aligned} & \text{minimize} \quad \sum_{i=1}^n w_i |x_i|, \\ & \text{subject to} \quad \mathbf{x}^T \mathbf{a} = c, \\ & \quad \quad \quad x_i \geq 0. \end{aligned}$$

Suppose $n = 2$, $w_1 = 1$, and $\mathbf{a} = [0.5 \ 1]^T$. What is the condition on w_2 such that the solution to the problem satisfies $x_2 = 0$ and $x_1 > 0$? [8 marks]

$$b) \quad \min \quad \sum_{i=1}^n w_i |x_i|$$

$$\text{subject to} \quad \mathbf{x}^T \mathbf{a} = c$$

$$\min \quad w_1 |x_1| + w_2 |x_2|$$

$$\text{subject to} \quad a_1 x_1 + a_2 x_2 = c$$

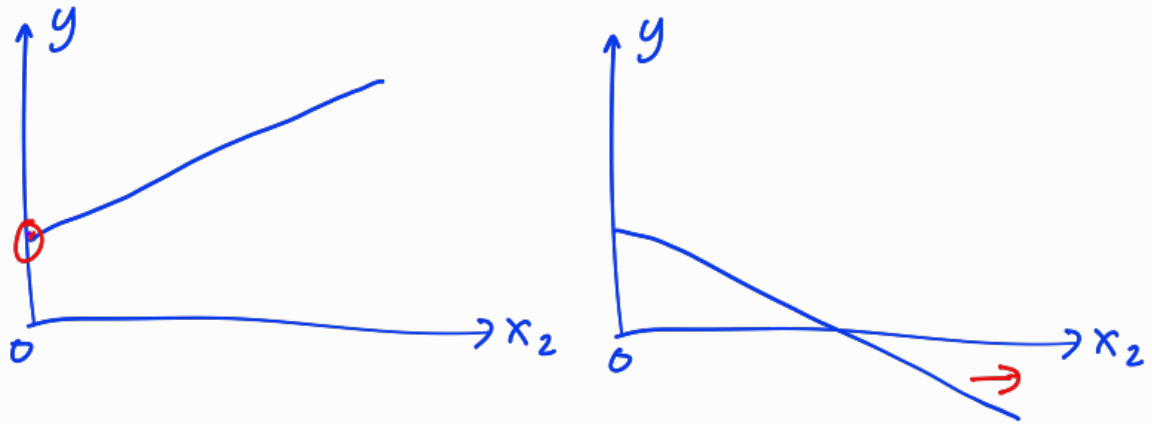
$$\min \quad x_1 + w_2 x_2 \quad x_1, x_2 \geq 0$$

$$\text{subject to} \quad 0.5 x_1 + x_2 = c$$

$$x_1 = 2(c - x_2)$$

$$\min 2(c - x_2) + w_2 x_2$$

$$\min 2c + (w_2 - 2)x_2$$



To minimize objective at $x_2 = 0$

$$w_2 - 2 > 0$$

$$w_2 > 2$$

$$x_1 = 2(c - 0) = 2c > 0$$

required condition $w_2 > 2$

(c) Suppose that Let $f(x) = h(g(x))$, where $g : \mathbb{R}^n \rightarrow \mathbb{R}$ is a convex function and $h : \mathbb{R} \rightarrow \mathbb{R}$ is a convex, non-decreasing function. Prove that f is convex. [7 marks]

$$f(x) = h(g(x))$$

$$g : \mathbb{R}^n \rightarrow \mathbb{R} \quad \text{convex}$$

$$h : \mathbb{R} \rightarrow \mathbb{R} \quad \text{convex, non decreasing}$$

Since g is convex,

$$g(\theta x + (1-\theta)y) \leq \theta g(x) + (1-\theta)g(y)$$

since h is non decreasing,

$$h(g(\theta x + (1-\theta)y)) \leq h(\theta g(x) + (1-\theta)g(y)) \quad \text{--- ①}$$

since h is convex,

$$h(\theta g(x) + (1-\theta)g(y)) \leq \theta h(g(x)) + (1-\theta)h(g(y)) \quad \text{--- ②}$$

①, ②

$$h(g(\theta x + (1-\theta)y)) \leq \theta h(g(x)) + (1-\theta)h(g(y))$$

$$f(x) = h(g(x))$$

$$f(\theta x + (1-\theta)y) \leq \theta f(x) + (1-\theta)f(y)$$

$f(x)$ is convex

(d) Consider following QCQP problem,

$$\begin{aligned} & \text{minimize } \frac{1}{2} \mathbf{x}^T \mathbf{P} \mathbf{x} + \mathbf{c}^T \mathbf{x} + d, \\ & \text{s.t. } \frac{1}{2} \mathbf{x}^T \mathbf{Q}_i \mathbf{x} + \mathbf{r}_i^T \mathbf{x} + s_i \leq 0, \quad i = 1, \dots, m, \\ & \mathbf{A} \mathbf{x} = \mathbf{b}, \\ & \text{with } \mathbf{P}, \mathbf{Q}_i \in \mathbb{S}_+^n. \end{aligned}$$

- i. What can you say about this problem, when $\mathbf{P}$ and $\mathbf{Q}_i$ are positive semi-definite? [2 marks]
- ii. If these matrices $\mathbf{P}$ and $\mathbf{Q}_i$ are neither positive nor negative semi-definite, what are the challenges arise when solving the QCQP problem? [3 marks]

① When $\mathbf{P}$ and $\mathbf{Q}_i$ are PSD,

objective function - convex Hessian is $\mathbf{P} \succeq 0$
quadratic constraints - convex Hessians are $\mathbf{Q}_i \succeq 0$
equality constraint - affine (convex)

QCQP is **convex** optimization problem.

- * local min = global min
- * polynomial time methods can solve reliably
- * strong duality under Slater's condition

② When $\mathbf{P}$ and $\mathbf{Q}_i$ are indefinite,

QCQP is **non convex** optimization problem.

- * multiple local minima
- * non convex feasible region
- * computationally difficult
- * non zero duality gap

Question 4.

- (a) A construction company builds three types of houses namely A, B, and C. On an area of 1000 m^2 , the company can build either three houses of type A, two houses of type B, or one house of type C. The company has a total of 50000 m^2 available, all of which must be used completely. Let x_A , x_B , and x_C denote the number of houses of types A, B, and C, respectively. The company wants to maximize its profit (in million) $P(x_A, x_B, x_C)$, given by

$$P(x_A, x_B, x_C) = x_A(x_A - 60) + 40 x_B + 30 x_C$$

Building houses of types A and C requires 1000 working hours each. Building a house of type B requires twice as many working hours as type A. A total of 10^5 working hours is available and must be fully used.

The construction costs for house type C are one million Rupees, which is twice the cost of house type A. The construction cost for house type B is three times that of house type C. The available budget is 105 million Rupees.

- Formulate the optimization problem for maximizing the profit as a function of x_A , x_B , and x_C in standard form. Also assume that the company must build all types of houses. [2 marks]
- Write down the KKT conditions including primal feasibility, dual feasibility, and complementary slackness. [3 marks]
- Solve the optimization problem using the KKT conditions and determine the values of x_A , x_B , and x_C that maximize the profit. (Hint: Neglect the constraint that x_A , x_B , and x_C must be non-negative.) [10 marks]
- The construction company decides to offer only one type of house. Which of the given house types should be chosen to maximize the profit? [5 marks]

$$i) \quad \max \quad x_A (x_A - 60) + 40 x_B + 30 x_C$$

$$\text{such that} \quad 1000 x_A + 2000 x_B + 1000 x_C = 100000$$

$$x_A + 2 x_B + x_C = 100 \quad \text{--- (1)}$$

$$333.3 x_A + 500 x_B + 1000 x_C = 50000$$

$$0.33 x_A + 0.5 x_B + x_C = 50 \quad \text{--- (2)}$$

$$0.5 x_A + 3 x_B + x_C \leq 105 \quad \text{--- (3)}$$

$$x_A > 0 \quad x_B > 0 \quad x_C > 0$$

$$\begin{aligned}
 \text{ii)} \quad L(x, \lambda, \mu) = & x_A (x_A - 60) + 40 x_B + 30 x_C + \\
 & \lambda_1 (x_A + 2 x_B + x_C - 100) + \\
 & \lambda_2 (0.33 x_A + 0.5 x_B + x_C - 50) - \\
 & \mu (0.5 x_A + 3 x_B + x_C - 105)
 \end{aligned}$$

stationarity

$$\frac{\partial L}{\partial x_A} = 0 \quad 2x_A - 60 + \lambda_1 + 0.33\lambda_2 - 0.5\mu = 0 \quad - (1)$$

$$\frac{\partial L}{\partial x_B} = 0 \quad 40 + 2\lambda_1 + 0.5\lambda_2 - 3\mu = 0 \quad - (2)$$

$$\frac{\partial L}{\partial x_C} = 0 \quad 30 + \lambda_1 + \lambda_2 - \mu = 0 \quad - (3)$$

primal feasibility

$$x_A + 2x_B + x_C = 100 \quad - (4)$$

$$0.33x_A + 0.5x_B + x_C = 50 \quad - (5)$$

$$0.5x_A + 3x_B + x_C \leq 105$$

dual feasibility $\mu \geq 0$

complementary slackness

$$\mu (0.5x_A + 3x_B + x_C - 105) = 0$$

$$0.5x_A + 3x_B + x_C - 105 = 0 \quad - (6) \quad , \mu > 0$$

$$\text{ii)} \quad (4), (5), (6) \Rightarrow x_A = 30, x_B = 20, x_C = 30$$

$$\begin{aligned} \text{max profit} \quad & x_A (x_A - 60) + 40 x_B + 30 x_C \\ & 30 (30 - 60) + 40 (20) + 30 (30) \\ & \underline{\underline{800 \text{ Million}}} \end{aligned}$$

$$\begin{aligned} \text{iv)} \quad & x_A + 2x_B + x_C \leq 100 && \text{hours} \\ & 0.33x_A + 0.5x_B + x_C \leq 50 && \text{land} \\ & 0.5x_A + 3x_B + x_C \leq 105 && \text{budget} \end{aligned}$$

A only

$$\text{max by hours} = 100$$

$$\text{max by land} = 150$$

$$\text{max by budget} = 210$$

can build 100 type A houses

$$\text{profit} = 100 (100 - 60) = 4000 \text{ Million}$$

B only

$$\text{max by hours} = 50$$

$$\text{max by land} = 100$$

$$\text{max by budget} = 35$$

can build 35 type B houses

$$\text{profit} = 40 \times 35 = 1400 \text{ Million}$$

C only

$$\text{max by hours} = 100$$

$$\text{max by land} = 50$$

$$\text{max by budget} = 105$$

Can build 50 type C houses

$$\text{profit} = 30 \times 50 = 1500 \text{ Million}$$

Building only type A houses give max profit.

(b) Given is the optimization problem:

$$\text{minimize } f(x, y) = (x - 2.4)^2 + (y - 1.5)^2$$

subject to:

$$2x + 3y \leq 6$$

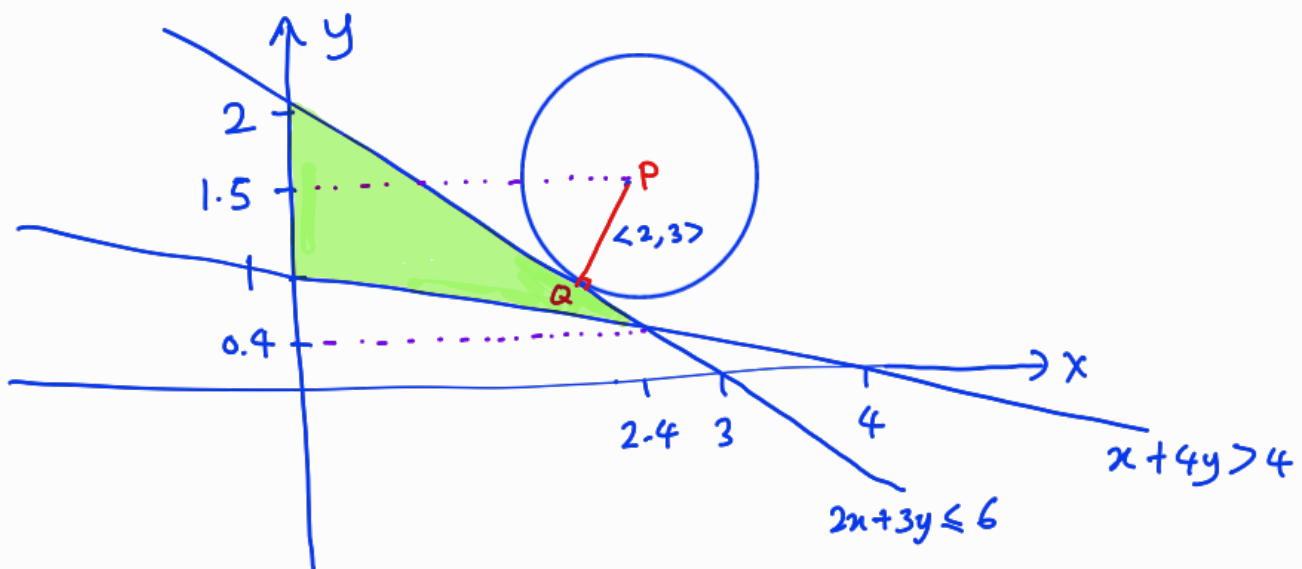
$$x + 4y > 4$$

$$x \geq 0$$

$$y \geq 0$$

Determine the minimum of $f(x, y)$ graphically.

[5 marks]



$$Q = P - t(2, 3)$$

$$Q = (2.4, 1.5) - t(2, 3)$$

$$Q = (2.4 - 2t, 1.5 - 3t)$$

Since Q is on $2x + 3y = 6$,

$$2(2.4 - 2t) + 3(1.5 - 3t) = 6$$

$$t = 0.253$$

$$Q = (2.4 - 2(0.253), 1.5 - 3(0.253))$$

$$Q = (1.892, 0.738)$$

check whether Q satisfies $x + 4y > 4$

$$1.892 + 4(0.738) > 4$$

$$4.844 > 4$$

$$f(x, y) = (1.892 - 2.4)^2 + (0.738 - 1.5)^2 = \underline{\underline{0.838}}$$

Question 5.

(a) Consider a convex optimization problem with m inequality constraints, formulated as:

$$\begin{aligned} & \underset{\mathbf{x}}{\text{minimize}} && f_0(\mathbf{x}), \\ & \text{subject to} && f_i(\mathbf{x}) \leq 0, \quad i = 1, \dots, m. \end{aligned}$$

Suppose that the Barrier method is used to solve this problem.

- Formulate the modified optimization problem using the Barrier method. [2 marks]
- State the main steps of the Barrier method. [3 marks]
- Prove that the solution of the barrier method converges to the Karush–Kuhn–Tucker (KKT) solution as the barrier parameter $t \rightarrow \infty$. [10 marks]
- The algorithm is initialized with a strictly feasible $\mathbf{x}$, an initial penalty parameter $t^{(0)} = 3$, a multiplier $\mu = 2$, and a termination tolerance $\epsilon = 10^{-3}$. For twenty inequality constraints ($m = 20$), determine the minimum number of the iterations. [5 marks]
- Discuss the impact of selecting low and high values of the parameter the multiplier μ on the computational complexity of the Barrier method. [5 marks]

i. minimize $f_0(\mathbf{x}) + \frac{1}{t} \phi(\mathbf{x})$

$$\phi(\mathbf{x}) = - \sum_{i=1}^m \ln(-f_i(\mathbf{x})) \quad \text{logarithmic barrier function}$$

$$\text{dom } \phi = \{ \mathbf{x} \in \mathbb{R}^n, f_i(\mathbf{x}) < 0 \}$$

ii. choose initial strictly feasible point $\mathbf{x}^0$ s.t. $f_i(\mathbf{x}^0) < 0$

select parameters $t = t_0, \mu > 1, \text{ tolerance } \epsilon > 0$

centering step $\mathbf{x}^*(t) = \min_{\mathbf{x}} [t f_0(\mathbf{x}) + \phi(\mathbf{x})]$

check stopping criteria $\frac{m}{t} < \epsilon$

increase barrier parameter $t \leftarrow \mu t$

repeat 3, 4, 5

iii. $x^*(t)$ - unique minimizer of the centering problem

$$F_t(x) = f_0(x) + \frac{1}{t} \phi(x)$$

$$\nabla F(x^*(t)) = 0$$

$$\nabla f_0(x^*(t)) + \frac{1}{t} \nabla \phi(x^*(t)) = 0$$

$$\nabla f_0(x^*(t)) + \sum_{i=1}^m \frac{1}{-t f_i(x^*(t))} \nabla f_i(x^*(t)) = 0$$

$$\nabla f_0(x^*(t)) + \sum_{i=1}^m \lambda_i^*(t) \nabla f_i(x^*(t)) = 0$$

$$\lambda_i^*(t) = \frac{-1}{t f_i(x^*(t))}$$

KKT conditions,

1) primal feasibility $f_i(x^*(t)) < 0$ satisfied

2) dual feasibility $f_i(x^*(t)) < 0, t > 0$
 $\frac{-1}{t f_i(x^*(t))} > 0$
 $\lambda_i^*(t) > 0$ satisfied

3) stationarity $\nabla f_0(x^*(t)) + \sum_{i=1}^m \lambda_i^*(t) \nabla f_i(x^*(t)) = 0$ satisfied

4) complementary slackness $\lambda_i^*(t) f_i(x^*(t)) = \frac{-1}{t f_i(x^*(t))} \cdot f_i(x^*(t)) = \frac{-1}{t}$

duality gap of
primal-dual pair, $\sum_{i=1}^m -\lambda_i^*(t) f_i(x^*(t)) = \frac{m}{t}$

$$t \rightarrow \infty, \quad \frac{m}{t} \rightarrow 0$$

$$\sum_{i=1}^m -\lambda_i^*(t) f_i(x^*(t)) \rightarrow 0 \quad \text{satisfied}$$

iv. algorithm terminates when,

duality gap $\leq$ tolerance

$$\frac{m}{t^k} \leq \varepsilon$$

$$\frac{m}{\mu^k t^0} \leq \varepsilon$$

$$t^k = \mu^k t^0$$

$$\mu^k \geq \frac{m}{t^0 \varepsilon}$$

$$k \geq \frac{\ln\left(\frac{m}{t^0 \varepsilon}\right)}{\ln(\mu)}$$

$$k \geq \frac{\ln\left(\frac{20}{3 \times 10^{-3}}\right)}{\ln(2)}$$

$$k \geq 12.7$$

$$k_{\min} = 13$$

minimum number of iterations = 13

v. Small μ

- many outer iterations
- stable convergence

large μ

- fewer outer iterations
- harder, unstable sub problems

Question 6.

- (a) The central path serves as a deterministic geometric trajectory through the interior of a feasible region. The logarithmic barrier function $\phi: \mathbb{R}^n \rightarrow \mathbb{R}$ is explicitly defined as:

$$\phi(\mathbf{x}) = - \sum_{i=1}^m \ln(-f_i(\mathbf{x})).$$

The implicit domain of this barrier function is strictly restricted to the interior of the feasible set:

$$\text{dom } \phi = \{\mathbf{x} \in \mathbb{R}^n \mid f_i(\mathbf{x}) < 0, i = 1, \dots, m\}.$$

- i. Mathematically demonstrate how the logarithmic barrier function $\phi(\mathbf{x}) = - \sum_{i=1}^m \ln(-f_i(\mathbf{x}))$ enforces strict feasibility, explaining why iterates can never violate or lie exactly on a constraint boundary. [5 marks]
- ii. Contrast the asymptotic behavior of the central path point $\mathbf{x}^*(t)$ under the two limiting cases: $t \rightarrow 0$ and $t \rightarrow \infty$. Identify the specific geometric points the trajectory targets in each case. [10 marks]

$$i) \quad \phi(\mathbf{x}) = - \sum_{i=1}^m \ln(-f_i(\mathbf{x}))$$

$$\text{dom } \phi = \{ \mathbf{x} \in \mathbb{R}^n, f_i(\mathbf{x}) < 0, i=1, 2, \dots, m \}$$

for every feasible interior point,

$$f_i(\mathbf{x}) < 0 \Rightarrow -f_i(\mathbf{x}) > 0$$

logarithmic is defined.

consider approaching the boundary,

$$f_i(\mathbf{x}) \rightarrow 0^-$$

$$-f_i(\mathbf{x}) \rightarrow 0^+$$

$$\ln(-f_i(\mathbf{x})) \rightarrow -\infty$$

$$-\ln(-f_i(\mathbf{x})) \rightarrow \infty$$

$$\phi(\mathbf{x}) \rightarrow \infty$$

objective becomes infinitely large near the boundary.

- $f_i(x) > 0$ iterates can't cross the boundary
- $f_i(x) = 0$ iterates can't lie exactly on the boundary
- $f_i(x) < 0$ iterates remain strictly feasible.

$\therefore$ logarithmic barrier enforces strict feasibility for all iterates.

ii) Central path problem,

$$x^*(t) = \arg \min_x (t f_0(x) + \phi(x))$$

$$\underline{t \rightarrow \infty}$$

$$t f_0(x) \gg \phi(x) \quad \text{objective function dominates}$$

$$x^*(t) \rightarrow \arg \min f_0(x) \quad f_i(x) \leq 0$$

$$x^*(t) \rightarrow x^* \quad \text{optimal solution of original problem}$$

$$\underline{t \rightarrow 0}$$

$$t f_0(x) \ll \phi(x) \quad \text{barrier term dominates}$$

$$x^*(t) \rightarrow \arg \min \phi(x)$$

$$x^*(t) \rightarrow x^{ac} \quad \text{analytic center of feasible region}$$

iii. Consider a convex optimization problem

$$\begin{aligned} & \underset{x}{\text{minimize}} && f_0(x) = c^T x \\ & \text{subject to} && f_i(x) \leq 0, \quad i = 1, \dots, m \\ & && Ax = b. \end{aligned}$$

Show that centrality condition is

$$tc + \sum_{i=1}^m \frac{1}{b_i - a_i^T x} a_i = tc + A^T d = 0.$$

[10 marks]

barrier objective

$$F_t(x) = t f_0(x) + \phi(x)$$

$$F_t(x) = t c^T x - \sum_{i=1}^m \ln(b_i - a_i^T x)$$

$$\nabla(t c^T x) = tc$$

$$\nabla[-\ln(b_i - a_i^T x)] = \frac{1}{b_i - a_i^T x} \cdot a_i$$

$$\nabla F_t(x) = tc + \sum_{i=1}^m \frac{1}{b_i - a_i^T x} a_i$$

At the minimizer,

$$\nabla F_t(x^*(t)) = 0$$

$$tc + \sum_{i=1}^m \frac{1}{b_i - a_i^T x} a_i = 0$$

$$tc + \sum_{i=1}^m d_i a_i = 0$$

$$tc + A^T d = 0$$

centrality condition

$$tc + \sum_{i=1}^m \frac{1}{b_i - a_i^T x} a_i = tc + A^T d = 0$$